PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

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Pick Your Fairness

Example: Employment

Hiring algorithm:

- 1000 applicants
- 700 male, 300 female
- Finds 500 to be qualified (300 male, 200 female)
- Recommends 100 for interviews

Initial Results

Algorithm Recommendation (I)

	Male applicants	Female applicants	Total
Total Applicants	700	300	1000
Unqualified Applicants	400	100	500
Qualified Applicants	300	200	500
Qualification Rate	42.9%	66.7%	50%
Selected For Interview	**75**	**25**	**100**
Rejection Rate (qualified applicants)	75%	87.5%	80%
Rejection Rate (all applicants)	89.3%	91.7%	90%

What's "Fair"?

Concepts:

- *Individualized assessment*
  - E.g., "blind rankings;" ignore group status
  - Potential benefits: optimality; legitimacy
  - Potential costs: depends critically on ranking process; differential impacts $\rightarrow$ propagate existing biases; false precision
- *Demographic parity*
  - Equal interview rates across groups
  - Potential benefits: Representational equality; perceived fairness
  - Potential costs: Ignores different rates of qualifications
- *Equal Opportunity*
  - *Qualified* individuals are equally likely to be interviewed irrespective of group membership
  - Potential benefits: Procedural equality; accounts for different qualification rates
  - Potential costs: $\rightarrow$ different interview rates by group; requires valid qualification decisions
- Others (e.g., *equalized odds*)

Demographic Parity

Algorithm Recommendation (II)

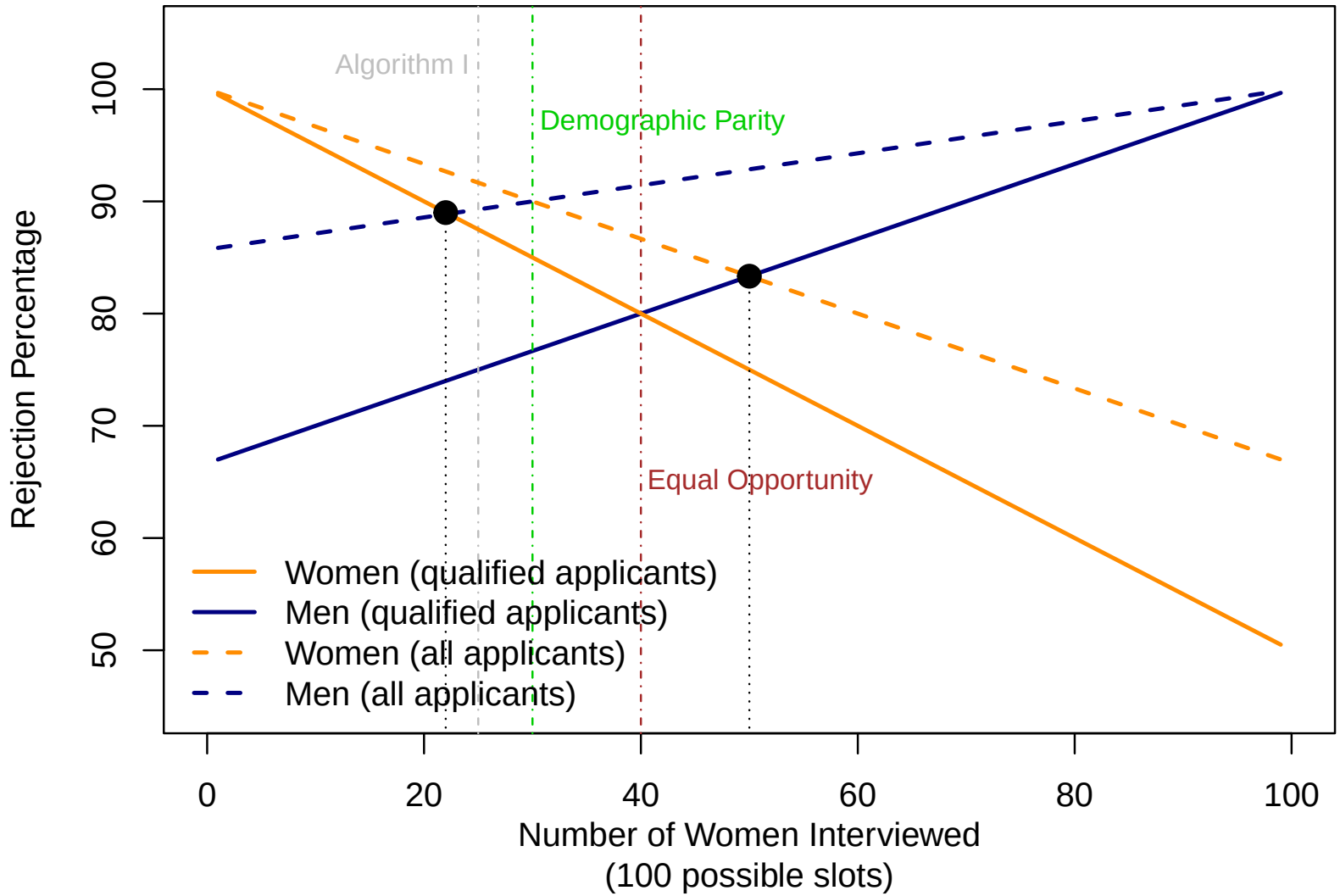
	Male applicants	Female applicants	Total
Total Applicants	700	300	1000
Unqualified Applicants	400	100	500
Qualified Applicants	300	200	500
Qualification Rate	42.9%	66.7%	50%
Selected For Interview	**70**	**30**	**100**
Rejection Rate (qualified applicants)	76.7%	85%	80%
Rejection Rate (all applicants)	90%	90%	90%

Equal Opportunity

Algorithm Recommendation (III)

	Male applicants	Female applicants	Total
Total Applicants	700	300	1000
Unqualified Applicants	400	100	500
Qualified Applicants	300	200	500
Qualification Rate	42.9%	66.7%	50%
Selected For Interview	**60**	**40**	**100**
Rejection Rate (qualified applicants)	80%	80%	80%
Rejection Rate (all applicants)	91.4%	86.7%	90%

Fairnesses, Illustrated



Fairness, Revisited

Pick your poison:

- Individualized assessment: Ideal in the absence of bias
- Demographic parity: Addresses structural inequalities
- Equal opportunity: "Meritocratic" (assuming qualification is unbiased)